Amrita School of Computing, Chennai

Course Delivery Plan

Course Code: Course Name	23CSE301: MACHINE LEARNING	**Program**	B. Tech – Computer Science and Engineering
L-T-P-C	3-0-2-4	**Semester / Year**	V/III
Name(s) of the Faculty	Dr. S. S. Chakravarthi, Dr. M. Rahul Raj	**Pre-requisite**	23MAT117 Linear Algebra, 23MAT216 Probability and Random Processes
Course Mentor	Dr. S. S. Chakravarthi	**Academic Year**	2026-27
Course Overview	The aim of this course is to provide foundational knowledge in machine learning. The students will learn to implement, train and validate the machine learning models and understand the recent algorithms in machine learning through case studies.		

<table>
<tr><th>Course Objectives</th><th colspan="2">Course Outcomes</th></tr>
<tr><td rowspan="5">Machine learning covers significant ground in various verticals – including image recognition, medicine, cyber security, facial recognition, business analytics etc. Study of ML lays a strong foundation for the study on NLP, Deep Learning, Reinforcement Learning, Graphical Models, AI, Predictive Analytics etc.

• The aim of this course is to provide foundational knowledge in machine learning.
• The students will learn to implement, train and validate the machine learning models and understand the recent algorithms in machine learning through case studies.</td><td>CO1</td><td>Understand the fundamental concepts, issues and challenges of machine learning.</td></tr>
<tr><td>CO2</td><td>Implement machine learning algorithms using programming tools and provide solution to real world applications.</td></tr>
<tr><td>CO3</td><td>Apply machine learning algorithms for parameter estimation or prediction problem.</td></tr>
<tr><td>CO4</td><td>Apply supervised learning techniques for classification problem and analyse their performance.</td></tr>
<tr><td>CO5</td><td>Apply un-supervised and ensemble learning techniques to solve a given problem.</td></tr>
</table>

Syllabus

Unit 1

Introduction to machine learning, Types of Machine learning, features, class boundary, Training, Validation & Testing, Generalization - underfit, regular fit and overfit; Loss /Cost function; Issues of Explainability, algorithmic bias, data and algorithmic privacy. Curse of Dimensionality, Dimensionality Reduction Techniques – Principal component analysis, Linear Discriminant Analysis, Feature selection – sequential & bi-directional. K-Nearest Neighbour classifier, Regression: Linear, logistic, Introduction to Regularization- LASSO, Ridge. Classifier Performance metrics – precision, recall, accuracy, f-score, AUC, Regression Performance metrics –RMSE, MAPE, R2 Score.

Unit 2

Supervised learning: Decision Trees, Support vector machines, Naive Bayes, Markov model, Hidden Markov Model, Artificial Neural Network - Perceptron & MLP with learning algorithms, Parameter Estimation: MLE and Bayesian Estimate, Expectation Maximization.

Unit 3

Unsupervised learning- Clustering – hierarchical & agglomerative, K means algorithm, cluster evaluation – Elbow technique; Ensemble learning – Bagging and boosting- Adaboost, Random Forest. Introduction to Reinforcement learning. Implementation of machine learning algorithms on real world problems and analyze their performance by tuning hyper parameters.

Textbooks / References

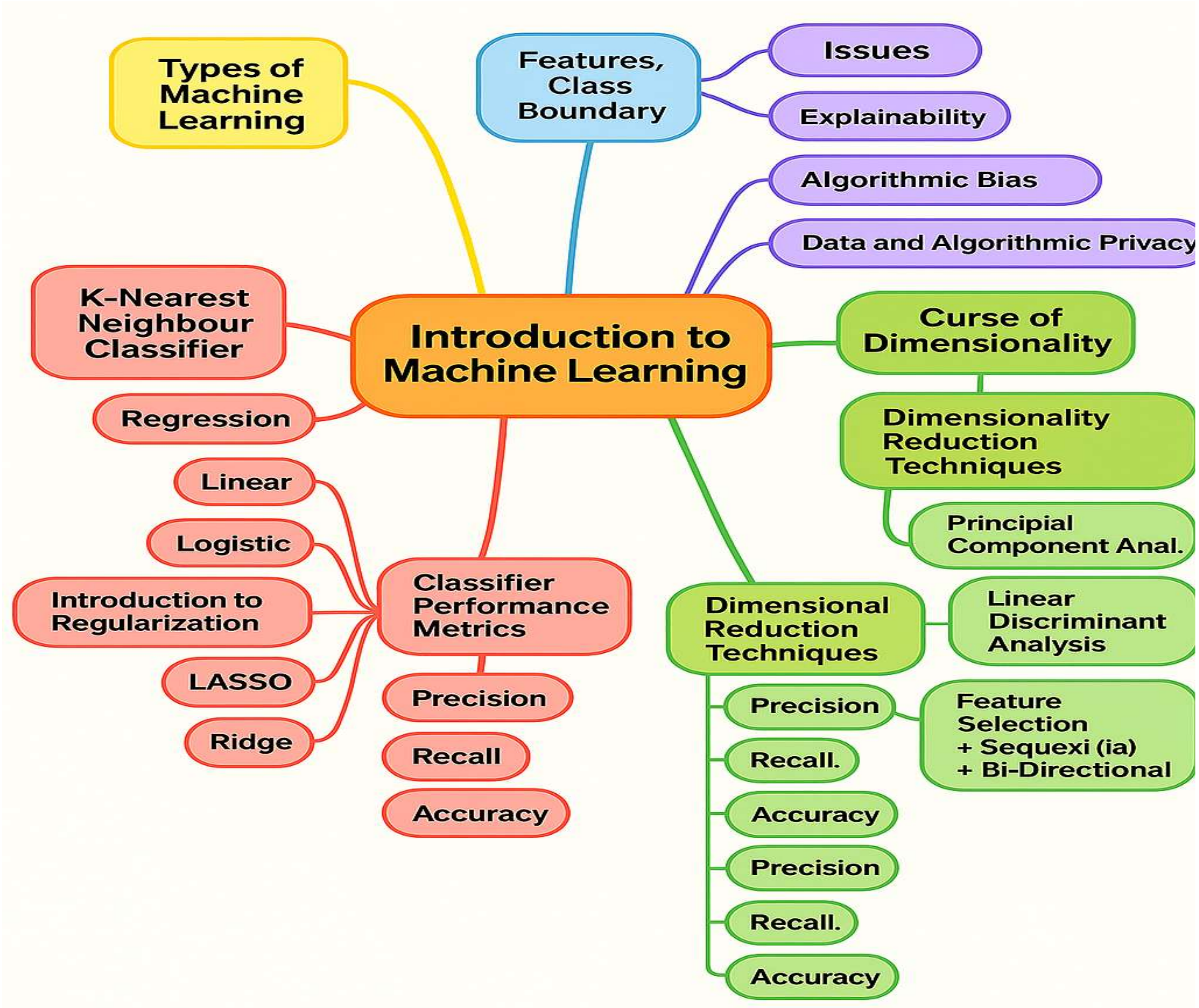
TEXTBOOK:

1. Tom Mitchell, "Machine Learning". McGraw Hill; ISBN: 9781259096952, 9781259096952; 2017.

REFERENCES:

1. Kevin P. Murphey. "Machine Learning", a probabilistic perspective. MIT Press, 2012.
2. Christopher M Bishop. "Pattern Recognition and Machine Learning". Springer 2010.
3. Richard O. Duda, Peter E. Hart, David G. Stork. "Pattern Classification". Wiley, Second Edition;2007.
4. Trevor Hastie, Robert Tibshirani, Jerome Friedman. "The Elements of Statistical Learning: Data Mining, Inference, and Prediction". Springer, 2017.

Concept Map



Types of Machine Learning
Features, Class Boundary
Issues
Explainability
Algorithmic Bias
Data and Algorithmic Privacy
K-Nearest Neighbour Classifier
Introduction to Machine Learning
Curse of Dimensionality
Regression
Dimensionality Reduction Techniques
Linear
Principial Component Anal.
Logistic
Classifier Performance Metrics
Introduction to Regularization
Dimensional Reduction Techniques
Linear Discriminant Analysis
LASSO
Precision
Precision
Feature Selection
+ Sequexi (ia)
+ Bi-Directional
Ridge
Recall
Recall.
Accuracy
Accuracy
Precision
Recall.
Accuracy

<table>
<tr><th colspan="5">Evaluation and Grading</th></tr>
<tr><th colspan="4">Internal (70)</th><th>External (30)</th><th>Total</th></tr>
<tr><th colspan="2">Components</th><th>Marks</th><th>Total Marks</th><td rowspan="8">End Semester
30</td><td rowspan="8">Internal + External
= 100</td></tr>
<tr><td colspan="2">Mid term</td><td>20</td><td>20</td></tr>
<tr><td>Continuous Assessment (Theory)</td><td>Assignments (3)</td><td>10</td><td rowspan="6">50</td></tr>
<tr><td rowspan="5">Capstone Project (Lab)</td><td>Lab Activities (6)</td><td>05</td></tr>
<tr><td>Foundation and Intermediate Level Certification in Machine Learning by Infosys Springboard</td><td>10</td></tr>
<tr><td>Phase – 1: Problem Statement Definition and Interpretation</td><td>05</td></tr>
<tr><td>Phase – 2: Features Selection and Learning Technique Identification</td><td>10</td></tr>
<tr><td>Phase – 3: Capstone Project Evaluation</td><td>10</td></tr>
</table>

Programme Outcome (PO)	
PO1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Ability to design and engineer, innovative, optimal and elegant computing solutions to interdisciplinary problems using standard practices, tools and technologies.
PSO2	Ability to learn emerging computing paradigms for research and innovation

CO – PO Affinity Map

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1	3	2	2	3									3	2
CO 2	3	2	3	2	3	1		3	3	3		3	3	3
CO 3	3	2	2	2									3	3
CO 4	3	2	2	2									3	3
CO 5	3	2	2	2									3	3

3 – Strong, 2 – Moderate, 1 – Weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	**Level 2**	**Level 3**
50%	**60%**	**50%**	**60%**	**70%**

Online Class	Topics to be covered	Mode of Teaching	In-Class Activities	Out- Class Student Activities (E – References)	CO Mapping	Reference
Unit- 1						
1	Introduction to Machine Learning, Types of Machine learning	PPT	Direct instruction method	https://www.youtube.com/watch?v=r4sgKrRL2Ys	C01	T1

2	Features, class boundary, Training, Validation & Testing	PPT	Direct instruction method	https://www.youtube.com/watch?v=dSCFk168vmo	C01	T1
3	Generalization - underfit, regular fit and overfit	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=o3DztvnfAJg	C01	T1
4-5 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 1, 2 as per the Lab Activity Book.		CO2	
6	Loss /Cost function	PPT	Discussion and Peer Learning	https://www.youtube.com/shorts/rQ4ABFsQqDc	C01	T1
7	Issues of Explainability algorithmic bias, data and algorithmic privacy	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=W21YA1dLb6o	C01	R1-R3
8	Curse of Dimensionality	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=_4DaqzLyT08	C02	R1-R3
9-10 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 3, 4 as per the Lab Activity Book.		CO2	
11	Dimensionality Reduction Techniques, Introduction to PCA	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=99SBxEDVfZo	C02	R1-R3
12	Regression: Linear Regression	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=_M7Km1XZERU	CO2	T1
13	Logistic Regression	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=g__LURKuIj4	CO1	T1
14-15 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 5, 6 as per the Lab Activity Book.		CO2	
16	Feature selection – sequential & bi-directional	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=GJnSBhxWEEg	C02	T1
17	K-Nearest Neighbour classifier	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=2ATqoglcHus	C02	T1
18	Principal component analysis	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=FdzK33RymyQ	C02	T1

19-20 (L)	Identification of Real-time Problem Statement for Research Paper	Lab Evaluation	Presentation and Evaluation		CO5	
21	Linear Discriminant Analysis	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=UudeDPTtMos	C02	T1
22	Introduction to Regularization	PPT	Peer Learning & Problem based Learning	https://www.youtube.com/watch?v=2bx9Os2gPu4	CO1	T1
23	LASSO Regression	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=9lRv01HDU0s	CO2	T1
24-25 (L)	Identification of Real-time Problem Statement for Research Paper	Lab Evaluation	Presentation and Evaluation		CO5	
26	Ridge Regression	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=9lRv01HDU0s	CO2	T1
27	Classifier Performance metrics: AUC Precision, Recall, Accuracy, F-Score	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=aWAnNHXIKww	CO1	T1
Assignment – 1: Identify & Explain a real-time research problem which are being solved with Machine Learning. Also, discuss the ML technique proposed.						
28	Revision of Unit 1	PPT	Discussion and Peer Learning		CO1	T1
29-30 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 9, 11 as per the Lab Activity Book.		CO2	
Unit – 2						
31	Supervised learning	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=OTAR0kT1swg	CO1	T1
32	Decision Trees	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=yG1nETGyW2E	CO2	T1
33	Support vector machines	PPT	Problem based Learning	https://www.youtube.com/watch?v=jkFjfdQ1JDQ	CO2	T1

34-35 (L)	Literature review on Topic, Method and Approach and Datasets	Lab Evaluation	Presentation and Evaluation		CO5	
36	Naive Bayes	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=1B_lrrEo5ds	CO3	T1
37	Markov model	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=i3AkTO9HLXo	CO3	R1-R3
38	Hidden Markov Model	PPT	Peer Learning & Problem based Learning	https://www.youtube.com/watch?v=qGtth60V6nM	CO3	R1-R3
39-40 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 7, 8 as per the Lab Activity Book.		CO2	
41	Artificial Neural Network - Perceptron	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=fUTwNOvoiyA	CO3	R1-R3
42	MLP with learning algorithms	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=_VXGGhRprAg	CO3	T1
43	MLP with learning algorithms	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=_VXGGhRprAg	CO3	T1
44-45 (L)	Identification of Methodology & Limitations	Lab Evaluation	Presentation and Evaluation		CO5	
46	Parameter Estimation: : MLE	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=EuyEmNHgskU	CO3	T1
47	Parameter Estimation: Bayesian Estimate,	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=00krscK7iBA	CO4	T1
48	Expectation Maximization	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=NgWiPBH6jHA	CO4	T1
49-50 (L)	Development of Training Phase	Lab Evaluation	Presentation and Evaluation		CO5	
Assignment – 2: For the identified real-time problem (or similar problem), conduct the literature survey on the possible ML solutions						

51	Revision of Unit II	PPT	Discussion and Peer Learning		CO4	T1
Unit - 3						
52	Unsupervised learning	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=HTSCbxSxs-g	CO3	R1-R3
53	Clustering- hierarchical & agglomerative	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=zop2zuwF_bc	CO3	R1-R3
54-55 (L)	Development of Training Phase	Lab Evaluation	Presentation and Evaluation		CO5	
56	Clustering- hierarchical & agglomerative	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=tlIv3IT_hHk	CO3	R1-R3
57	K means algorithm	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=z7mlHoV1enk	CO4	R1-R3
58	Cluster evaluation – Elbow technique	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=wW1tgWtkj4I	CO4	R1-R3
59-60 (L)	Machine Learning Hands-on Experimentation	Lab Activity	Work on Experiment – 10, 12 as per the Lab Activity Book.		CO2	
61	Ensemble learning – Bagging	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=foWzsWFAmas	CO4	R1-R3
62	Ensemble learning - Boosting-Adaboost	PPT	Problem based Learning	https://www.youtube.com/watch?v=NrdtKndsC1I	CO4	R1-R3
Assignment – 3: Discuss the datasets and their features that can be devised for proposing the ML solution to the identified real-time problem.						
63	Ensemble learning - Adaboost	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=foWzsWFAmas	CO5	R1-R3
64-65 (L)	Hyper tuning the Model	Lab Evaluation	Presentation and Evaluation		CO5	
66	Random Forest	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=6K48CbOm99Y	CO5	R1-R3

67	Introduction to Reinforcement learning	PPT	Discussion and Peer Learning	https://www.youtube.com/watch?v=9vMpHk44XXo	CO5	T1
68	Revision of Unit III	PPT	Discussion and Peer Learning		CO5	T1
69-70 (L)	Capstone Project	Lab Evaluation	**Capstone Project Presentation and Evaluation**		CO5	
71	Case Study: Recommendation systems – I	Live Demo and Interaction	Project based & Peer Learning	https://www.youtube.com/watch?v=n3RKsY2H-NE	CO5	T1
72	Case Study: Recommendation systems – II	Live Demo and Interaction	Project based & Peer Learning	https://www.youtube.com/watch?v=n3RKsY2H-NE	CO5	T1
73	Case Study: Recommendation systems – III	Live Demo and Interaction	Project based & Peer Learning	https://www.youtube.com/watch?v=n3RKsY2H-NE	CO5	T1
74-75 (L)	Capstone Project	Lab Evaluation	**Capstone Project Presentation and Evaluation**		CO5	
End Semester Examination						

Assignment Evaluation Rubrics:

Assignment: (50 Marks)			
Evaluation Component	**Excellent (3)**	**Good (2)**	**Not Impressive (1)**
Assignment Content & Presentation **(30 Marks)**	The student incorporated all the suggestions given during the Lab Evaluation **(30x1=30 Marks)**	The student incorporated 50% of the suggestions given during the Lab Evaluation **(30x0.7=21 Marks)**	The student incorporated 20% to 30% of the suggestions given during the Lab Evaluation **(30x0.5=15 Marks)**
Plagiarism & AI Tools Usage **(20 Marks)**	The similarity index of the assignment is less than 10% with 30% AI Tools usage **(20x1=20 Marks)**	The similarity index of the assignment is between 10% to 30% with 30% to 45% AI Tools usage **(20x0.7=14 Marks)**	The similarity index of the assignment is More than 30% with higher than 45% AI Tools usage **(20x0.5=10 Marks)**

Lab Evaluation Rubrics:

Lab Evaluation – 1: Problem Statement Definition and Interpretation (100 Marks)			
Evaluation Component	**Excellent (1)**	**Good (0.6)**	**Not Impressive (0.4)**
Selection of the Problem **(30 Marks)**	The problem is real time and has scope to work **(30x1=30 Marks)**	The problem is not a real-time but has scope to work **(30x0.6=18 Marks)**	The problem maybe a real-time but has limited scope to work **(30x0.4=12 Marks)**
Understanding of the Problem **(30 Marks)**	The student has in-depth understanding of the problem and has clarity on how to proceed with the "***todo***" work **(30x1=30 Marks)**	The student has understanding of the problem, but need more clarity on how to proceed with the "***todo***" work **(30x0.6=18 Marks)**	The student has minimal understanding of the problem and absolutely no idea on how to proceed with the "***todo***" work **(30x0.4=12 Marks)**
Identifying the datasets used and Interpretation of the Solution given on the dataset **(30 Marks)**	The student has clearly identified the dataset used and the ML technique proposed in the base paper **(30x1=30 Marks)**	The student has identified the dataset used and the ML technique proposed in the base paper, but lacks deep understanding & clarity **(30x0.6=18 Marks)**	The student has not identified the dataset used and no clarity on the ML technique proposed in the base paper **(30x0.4=12 Marks)**
Presentation & Interaction **(10 Marks)**	Student delivered his content clearly orally & verbally **(10x1=10 Marks)**	Student delivered his content orally & verbally, but with minimal errors **(10x0.6=06 Marks)**	Student delivered his content orally & verbally, but with blunders **(10x0.4=04 Marks)**

Lab Evaluation – 2: Features Selection and Learning algorithm / technique identification. (100 Marks)			
Evaluation Component	**Excellent (1)**	**Good (0.6)**	**Not Impressive (0.4)**
Feature selection **(30 Marks)**	The student has clearly identified the features to be used for the identified problem **(30x1=30 Marks)**	The student identified the features but not suitable to use for the identified problem **(30x0.6=18 Marks)**	The student has not identified the features for the identified problem **(30x0.4=12 Marks)**
Identification of existing Algorithm through Literature Survey **(30 Marks)**	The student has clearly identified the existing ML techniques proposed till date for the identified problem, through an extensive literature survey **(30x1=30 Marks)**	The student has identified the existing ML techniques proposed for the identified problem, but through moderate level of literature survey. **(30x0.6=18 Marks)**	The student has limited knowledge on existing ML techniques proposed for the identified problem, as his literature survey was very narrow **(30x0.4=12 Marks)**
Identification of Learning algorithm / technique / method **(30 Marks)**	The student has identified the learning technique / method suitable for the proposed work with appropriate justification **(30x1=30 Marks)**	The student has identified the learning technique / method suitable for the proposed work with non-convincing justification **(30x0.6=18 Marks)**	The student has not identified the learning technique / method and not inclined towards solving the problem **(30x0.4=12 Marks)**
Presentation & Interaction **(10 Marks)**	Student delivered his content clearly orally & verbally **(10x1=10 Marks)**	Student delivered his content orally & verbally, but with minimal errors **(10x0.6=06 Marks)**	Student delivered his content orally & verbally, but with blunders **(10x0.4=04 Marks)**

Capstone Project (150 Marks)			
Evaluation Component	**Excellent (3)**	**Good (2)**	**Not Impressive (1)**
Problem Identification and Literature survey **(40 Marks)**	The student has in-depth understanding of the problem and has highlighted gaps in existing literature and areas for future research. **(40x1=40 Marks)**	The student understanding of the problem, but lacks in highlighting the gaps in existing literature and areas for future research. **(40x0.6=24 Marks)**	The student has minimal understanding of the problem and has no idea of identifying research gaps in existing literature. **(40x0.4=16 Marks)**
Handling datasets and Data Preprocessing **(40 Marks)**	The student has Clearly identified data sources, addressed the inconsistencies in data and Justified for the extraction of specific features. **(40x1=40 Marks)**	The student has identified data sources but lacks in addressing the inconsistencies in data and Justification for the extraction of specific features. **(40x0.6=24 Marks)**	The student has identified data sources, but could not address the data inconsistencies and inappropriate justification on features extraction. **(40x0.4=16 Marks)**
Algorithmic Design & Approach **(40 Marks)**	The student has commendable knowledge on the proposed algorithm and produces convincing results for various test cases **(40x1=40 Marks)**	The student just has knowledge on the proposed algorithm, with limited results or no improvement on the benchmarked results. **(40x0.6=24 Marks)**	The student has very minimal knowledge on the proposed algorithm and no convincing results for various test cases. **(40x0.4=16 Marks)**
Presentation & Interaction **(30 Mark)**	Student delivered his content clearly orally & verbally **(30x1=30 Marks)**	Student delivered his content orally & verbally, but with minimal errors **(30x0.6=18 Marks)**	Student delivered his content orally & verbally, but with blunders **(30x0.4=12 Marks)**

Note: Every student has to select ONE paper from any of the domains from the given list (will be provided by 20th June, 2026)

Attendance Requirement:

(i) All students must maintain a **minimum of 75% attendance** in each course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.

(ii) Students who **fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams**. Requests for exceptions will not be considered under any circumstances.

(iii) Attendance will be calculated up to **three days prior to the commencement of the mid-term and end-semester exams**, as per the academic calendar.

Faculty **Course Mentor** **Chairperson**